



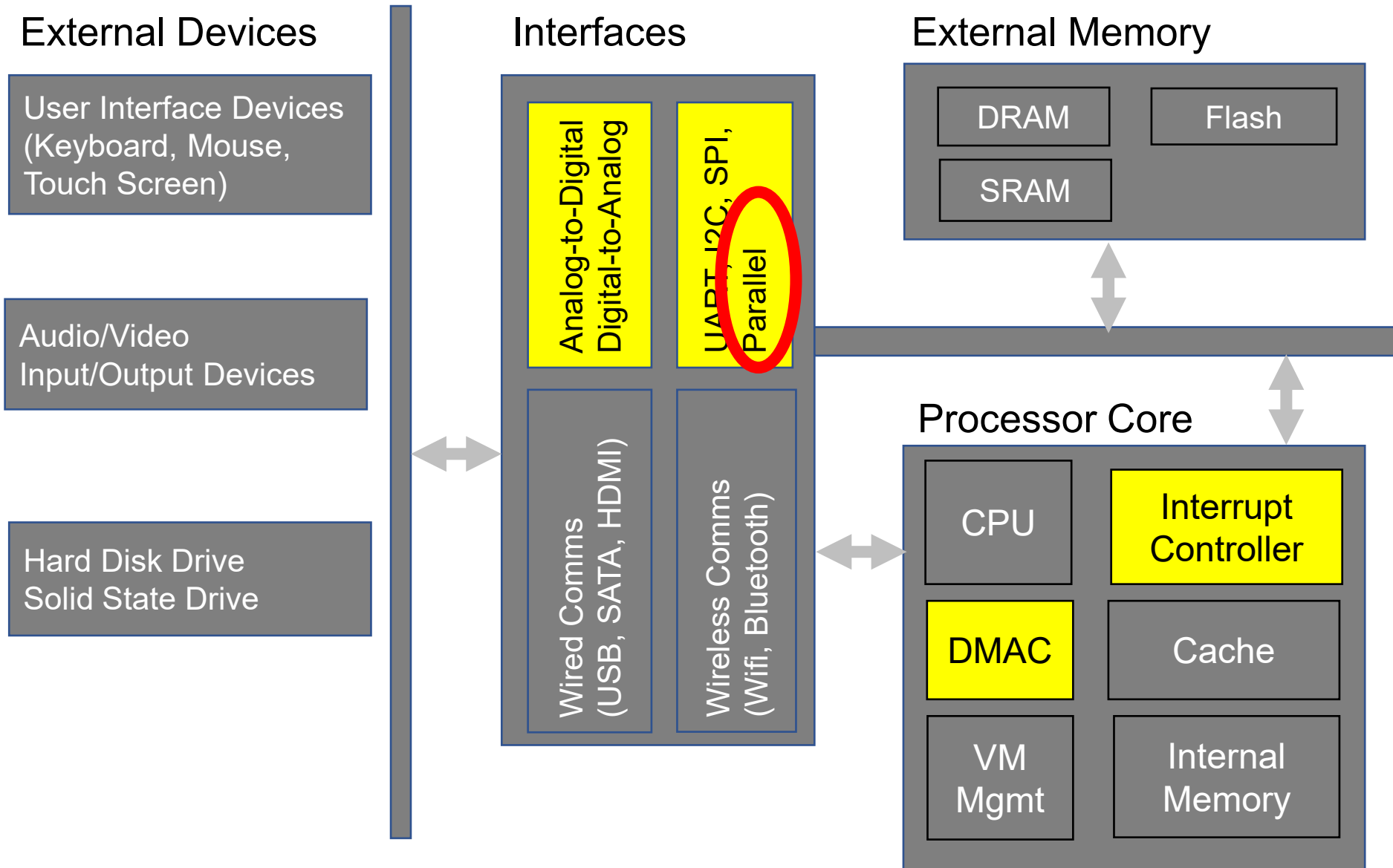
SC1006

Computer Organisation and Architecture

Parallel Data Transfer

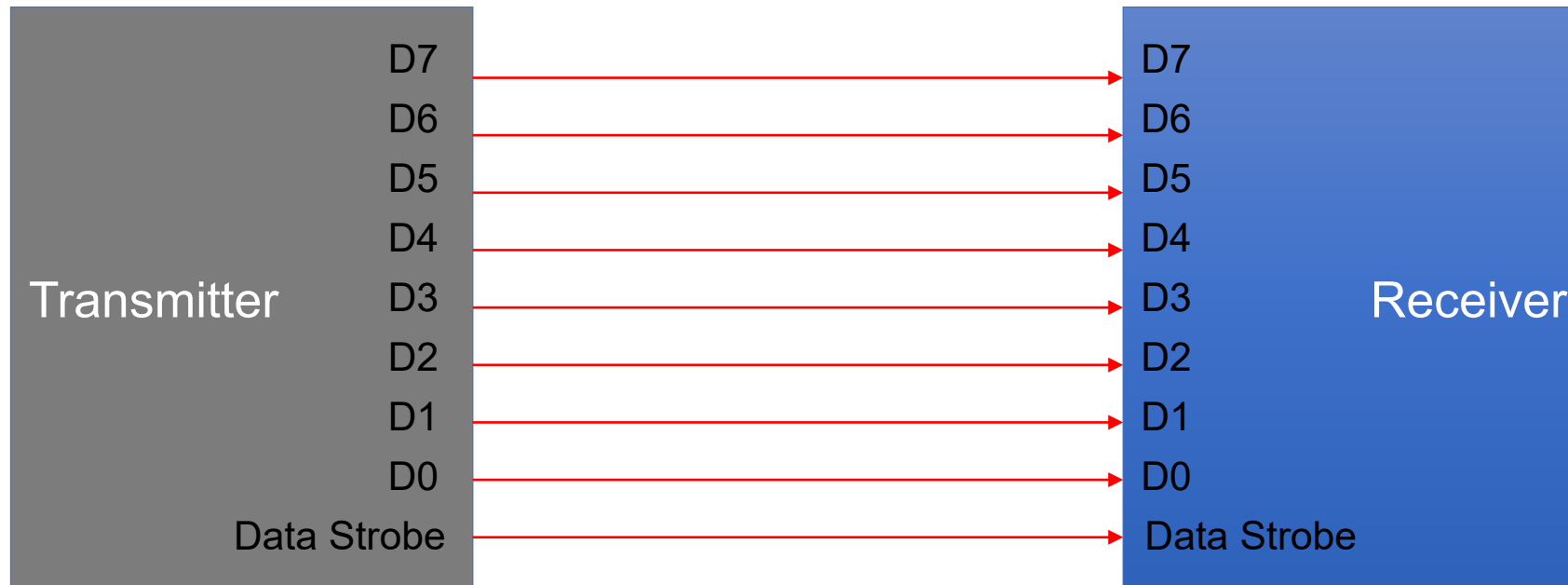
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Computer System Block Diagram



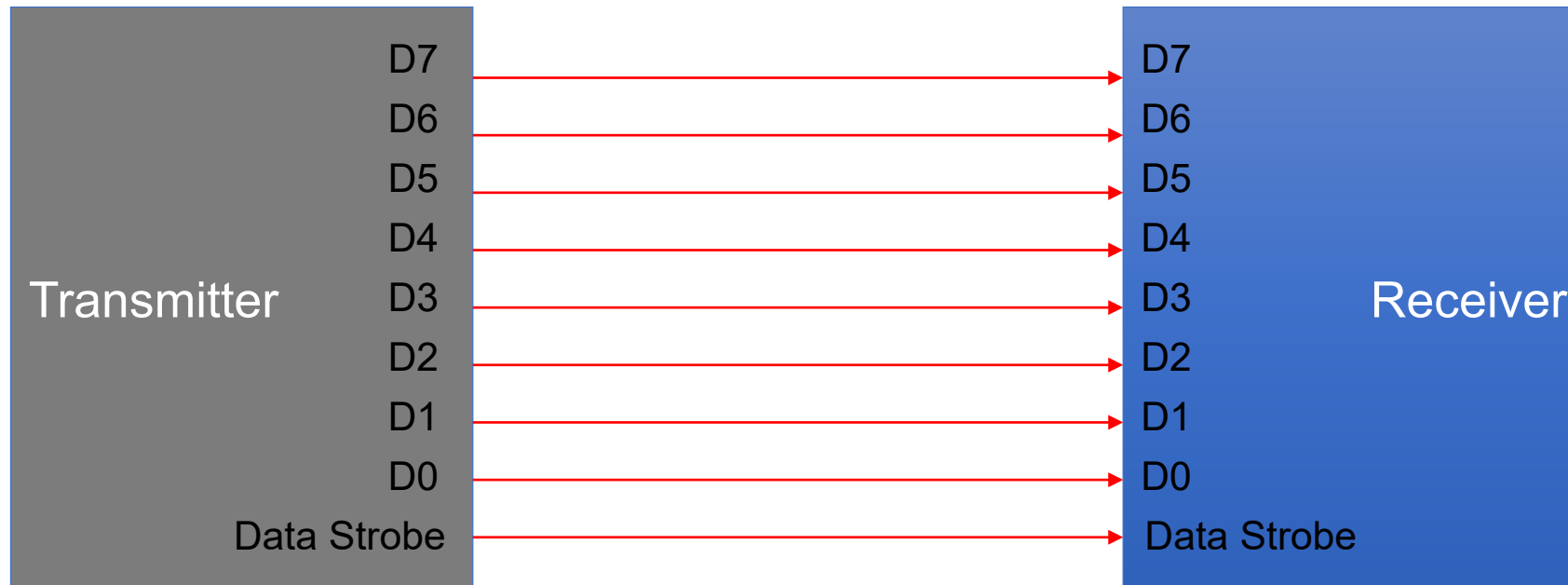
Parallel Data Transfer

- Multiple bits of data are transferred **simultaneously** between two devices.
- **Synchronous** in nature as some sort of **strobe signal** is needed to inform the receiver when to latch in the data. E.g. rising edge of strobe signal.



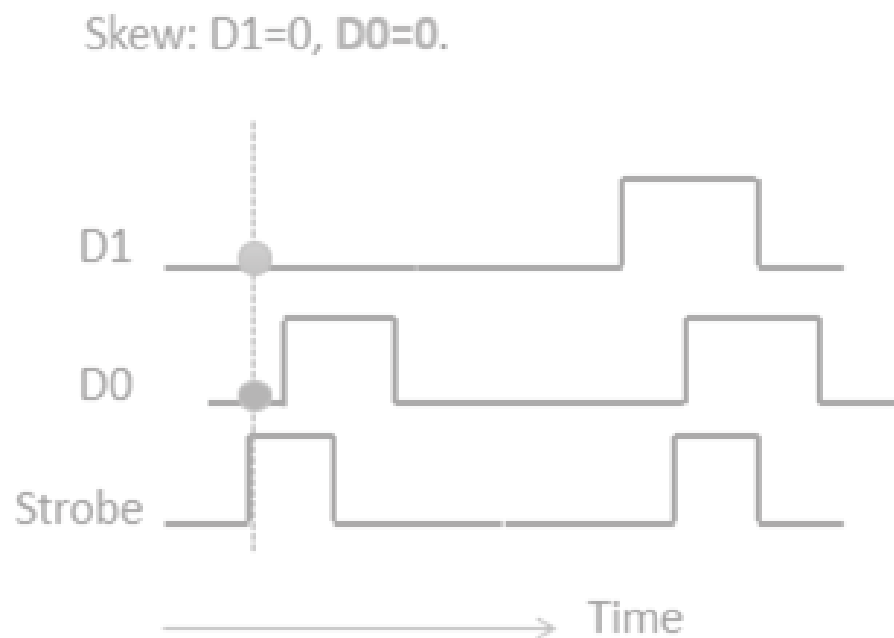
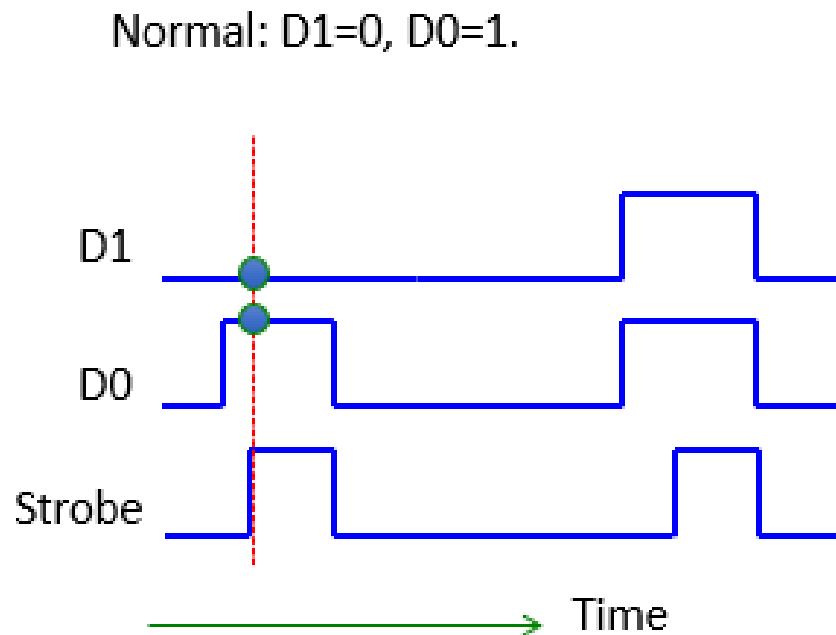
Parallel Data Transfer

- Multiple bits of data are transferred simultaneously between two devices.
- Synchronous in nature as some sort of strobe signal is needed to inform the receiver when to latch in the data. E.g. rising edge of strobe signal.
- Able to achieve **higher transfer rate** than Serial Interface (using same clock).
- But more prone to **Signal Skew** and **Crosstalk** (see later slides).



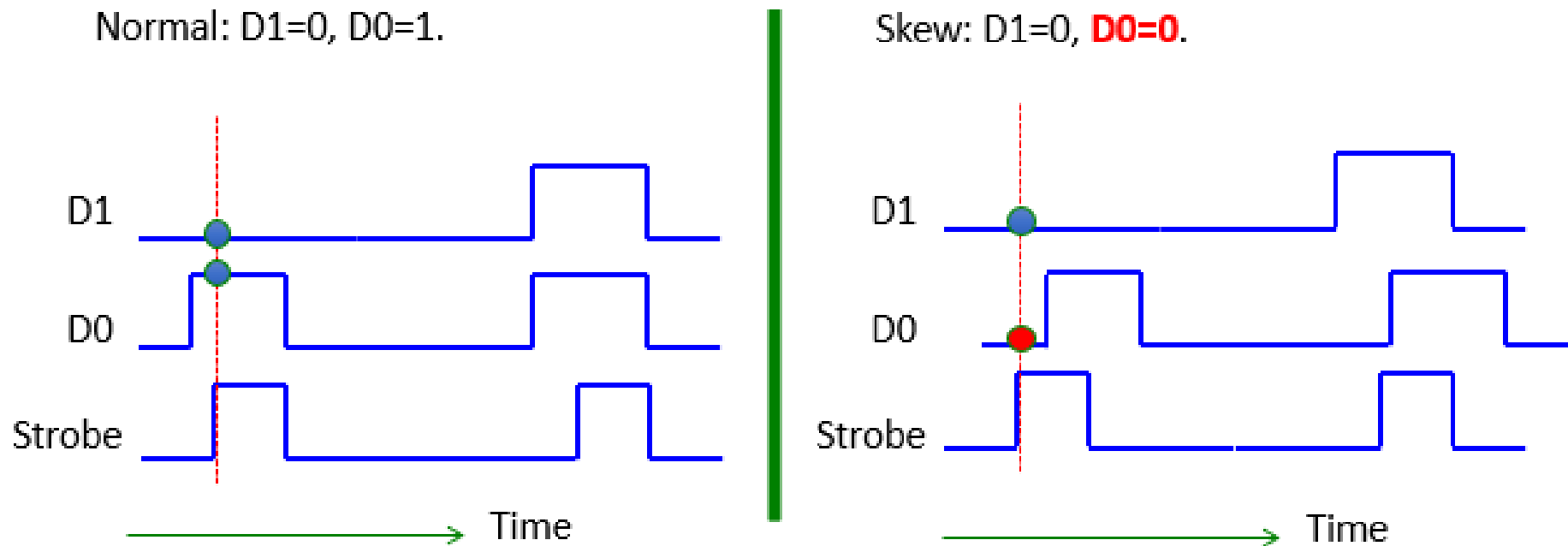
Signal Skew

- If for some reason (due to circuit design, external interference), the signal in one or more data lines **took different amount of time to reach the receiver**, then there is a skew between the signals in the parallel data bus.
- Below example illustrates the effect of signal skew. **Data is latched by the receiver on rising edge** of the strobe signal.
- This result in wrong data (D0=0 instead of 1) being latched by the receiver.



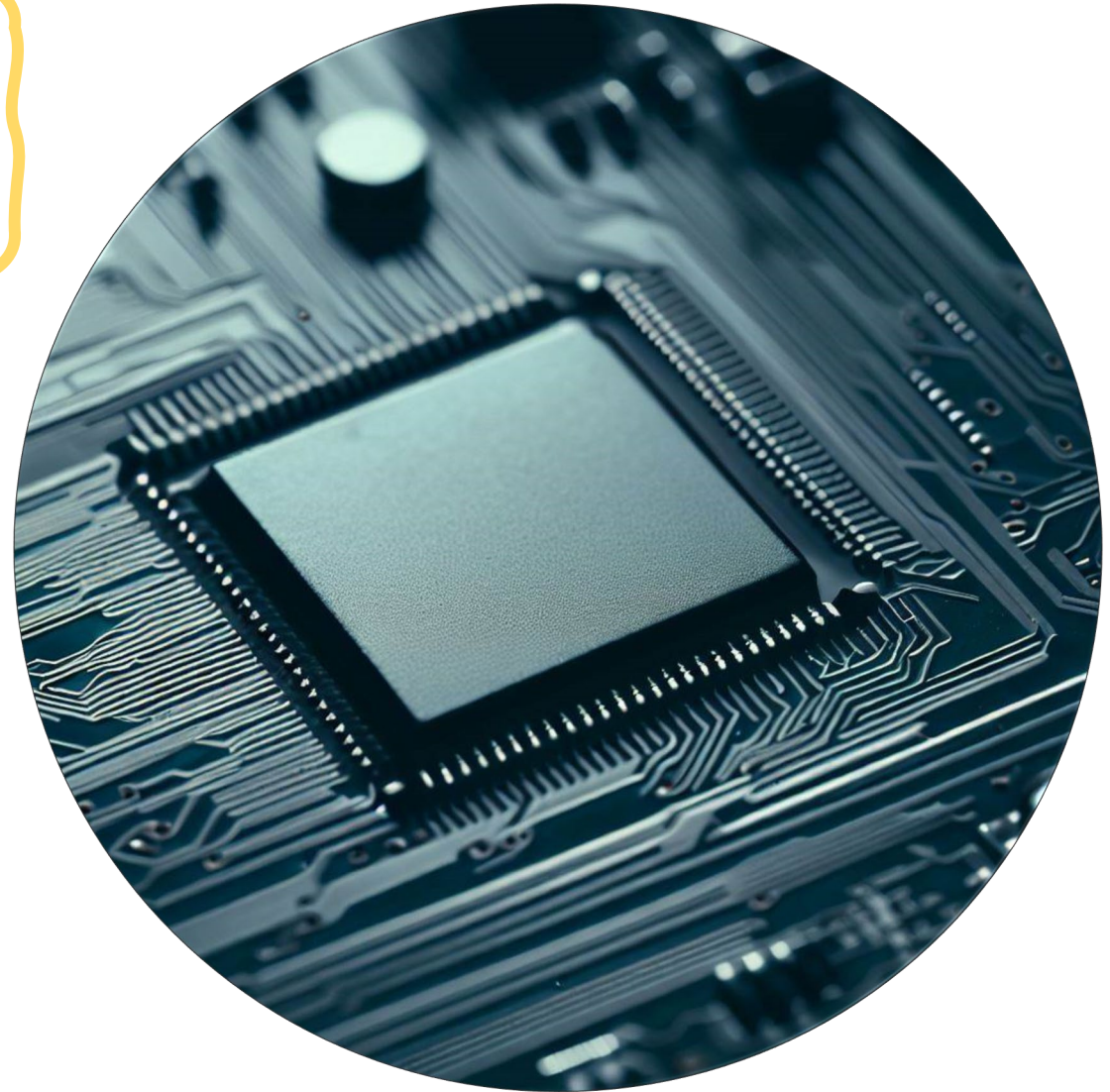
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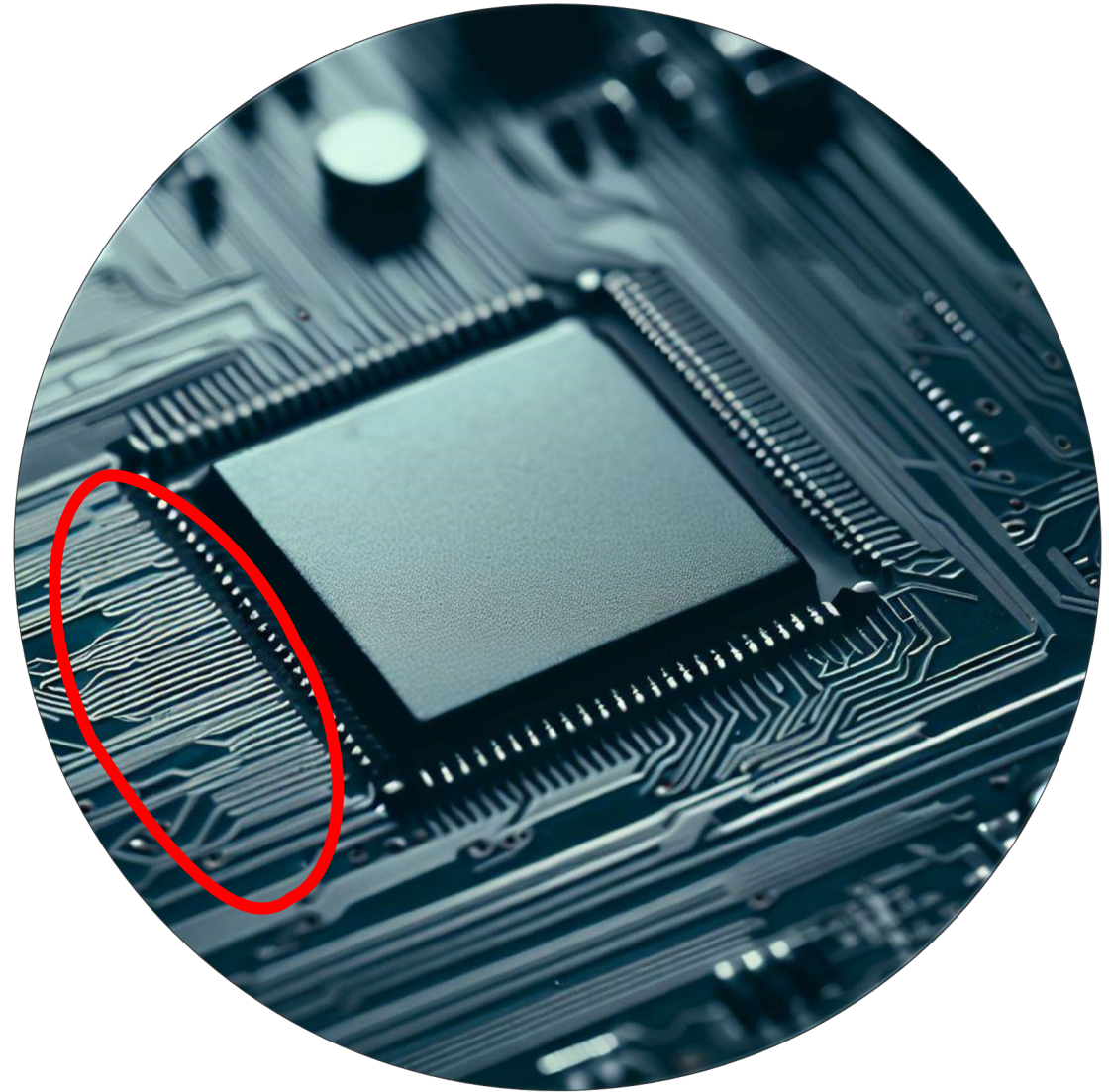
Signal Skew – Contributing Factors

- Signal Skew is due to **variation in propagation delay** between signals from the same data bus.
- Propagation delay is the **time taken for signal to travel between two points** in a circuit.
- **Capacitance** and **Resistance** of the physical data line is a major contributor to circuit propagation delay.
 - The larger the resistance and capacitance, the larger the delay. Illustrated in the equation $\tau = RC$ where τ is the time constant dictating rate of change of voltage levels in the data line concerned.
- Variation in resistance and capacitance of the signal lines can be due to
 - Variation in **PCB trace length/width** (lead to change in impedance).
 - Connecting **active components** (capacitors, inductors, IC etc) to some of the signal lines.



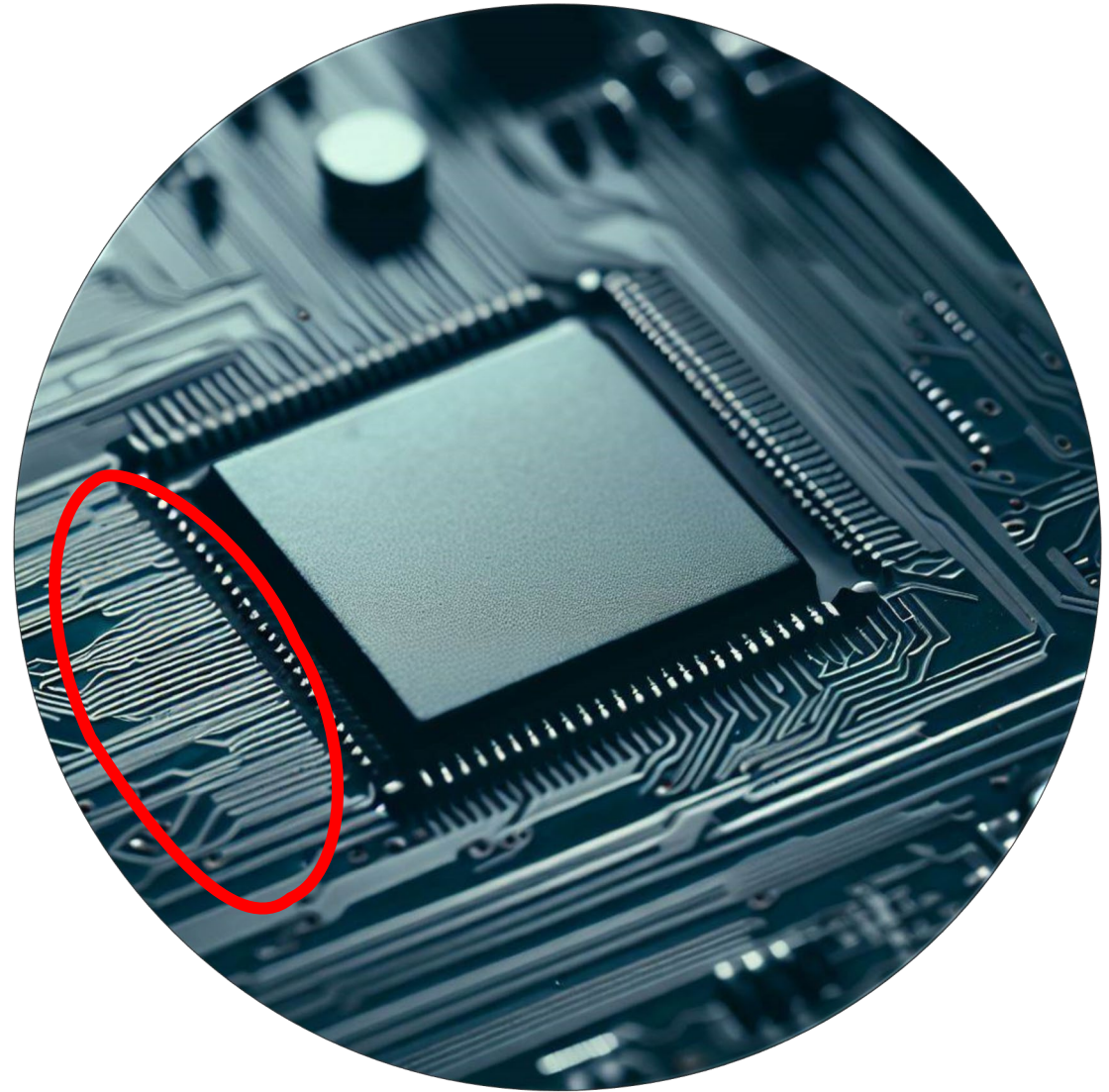
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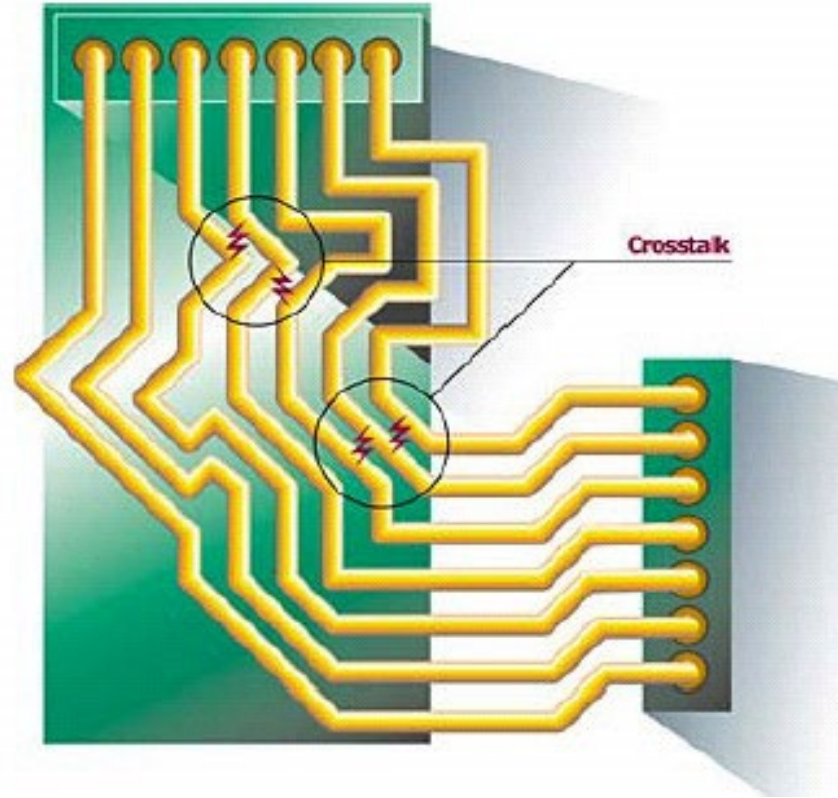


Crosstalk

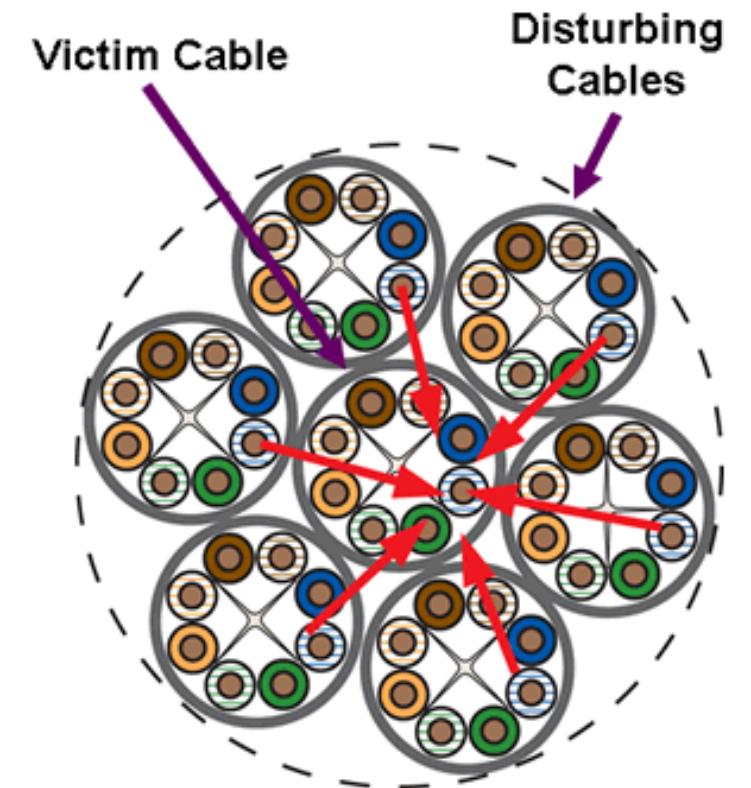
Party



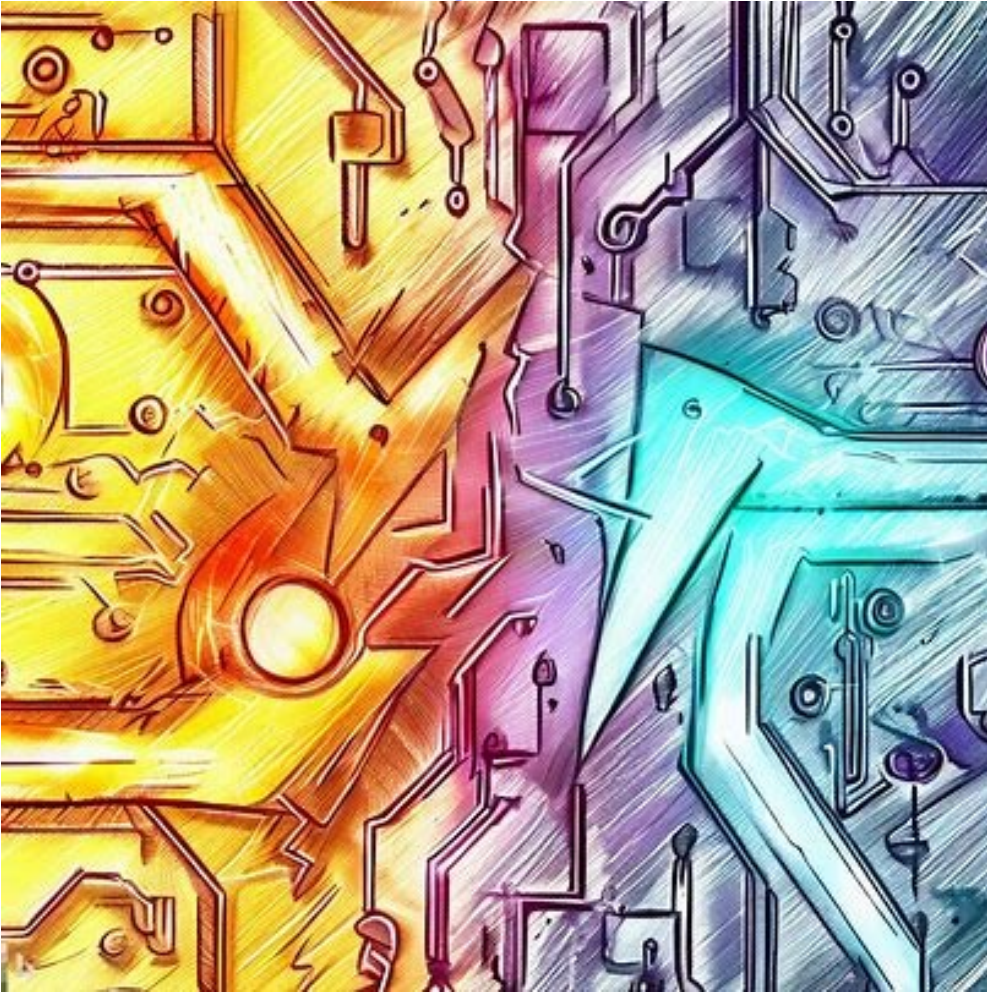
PCB



Cable



Crosstalk – Electrical Circuit



- Crosstalk are **undesired coupling of signals** from one circuit to another circuit.
- In a parallel bus context, the **close placement** of the data lines in PCB routing or cabling enables the effect of electrical signal in one trace/wire to be coupled over to the other. Creating **undesired interference** (crosstalk).
- Crosstalk can be transmitted **electrically** or via **electromagnetic radiation** (the trace/wire acts like an antenna).

Parallel Data Transfer – Pros and Cons

- Advantages
 - Fast data transfer rate (more bits can be transferred at one time)
- Disadvantages
 - Affected by **Signal Skew** and **Cross Talk**, which limits the maximum clocking speed and transfer distance.
 - **Higher hardware cost** compared to Serial data implementation.
 - Hardware (data cable) can be **bulky** if data width is large.
 - **Need more space** to route the PCB traces.

